NARAYANA IIT ACADEMY INDIA

Sec: SR.IIT_*CO-SC(MODEL-A,B&C) Date:27-01-2024

Time: 3HRS Max. Marks: 300

Name of the Student: ___________________ H.T. NO:

27 .01.24_SR.STAR CO-SC(MODEL-A,B&C)_Jee_Main__Jee-Main-GTM_28(N)_QP FINAL

PHYSICS: Extra Syllabus

CHEMISTRY: Extra Syllabus

MATHEMATICS: Extra Syllabus

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Match List I with List II:

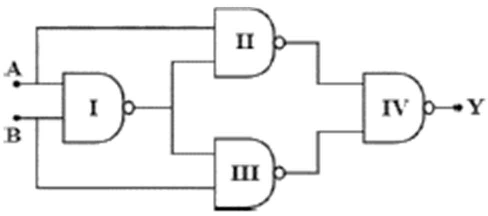
List - I		List - II	
A	Gauss's Law in electrostatics	I	$\oint \vec{E}\cdot d\vec{l} = -\frac{d\phi_B}{dt}$
B	Faraday's Law	II	$\oint \vec{B}\cdot d\vec{A} = 0$
C	Gauss's Law in Magnetism	III	$\oint \vec{B}\cdot d\vec{l} = \mu_0 i l C + \mu_0 \in_0 \frac{d\phi_E}{dt}$
D	Ampere-Maxwell Law	IV	$\oint \vec{E}\cdot d\vec{s} = \frac{q}{\epsilon_0}$

Choose the correct answer from the options given below:

1) $A - IV, B - I, C - II, D - III$ 2) $A - I, B - II, C - III, D - IV$

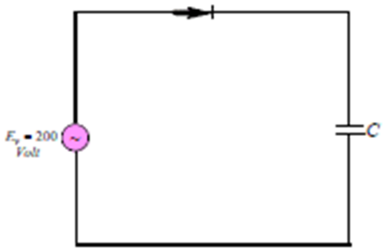
3) $A - III, B - IV, C - I, D - II$ 4) $A - II, B - IV, C - I, D - III$

2. Select the output Y of the combination of gates, for three sets of inputs $A = 1, B = 0; A = 1, B = 1$ and $A = 0, B = 0$ respectively:



1) $(0,1,1)$ 2) $(0,0,1)$ 3) $(1,0,0)$ 4) $(1,1,1)$

3. A sinusoidal voltage of r.m.s value of 200 volt is connected to the diode and capacitor C in the circuit, so that half wave rectifications occurs. The final potential difference in volt across C is:



1)500 2)200 3)283 4)141

4. A solid body of constant heat capacity $1\ J/{}^{0}C$ is being heated by keeping it in contact with reservoirs in two ways:

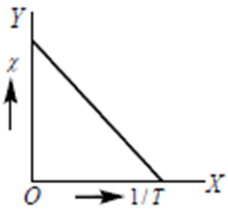
i) Sequentially keeping in contact with 2 reservoirs such that each reservoir supplies same amount of heat.

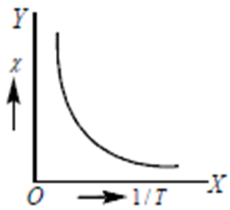
ii) Sequentially keeping in contact with 8 reservoirs such that each reservoir supplies same amount of heat.

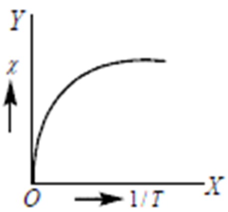
In both the cases body is brought from initial temperature $100^{\circ}C$ to final temperature $200^{\circ}C$. Entropy change of the body in the two cases respectively is:

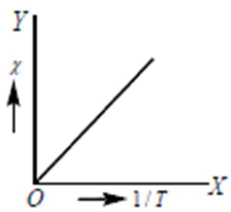
1) $\ln 2, 2\ln 2$ 2) $2\ln 2, 8\ln 2$ 3) $\ln 2, 4\ln 2$ 4) $\ln 2, \ln 2$

5. The graph between susceptibility χ and $1/T$ for paramagnetic material will be represented by, ($T =$ Absolute temperature)

1) 

2) 

3) 

4) 

6. Given below are two statements:

Statement I : If the Brewster's angle for the light propagating from air to glass is θ_B, then Brewster's angle for the light propagating from glass to air is $\frac{\pi}{2}-\theta_B$

Statement II : The Brewster's angle for the light propagating from glass to air is $\tan^{-1}(\mu_g)$ where μ_g is the refractive index of glass.

In the light of the above statements, choose the correct answer from the options given below:

1)Both statements I and II are true

2)Statement I is true but statement II is falseD

3)Both Statement I and II are false

4)Statement I is false but statement II is true

7. If the polarizing angle of a piece of glass for green light is 54.74°, then the angle of minimum deviation for an equilateral prism made of same glass is

(Given: $\tan 54.74^{0} = 1.414$)

1) 45° 2) 54.74^{0} 3) 60° 4) 30°

8. Two tubes of radii r_1 and r_2, and lengths l_1 and l_2, respectively, are connected in series and a liquid flows through each of them in streamline conditions. P_1 and P_2 are pressure differences across the two tubes. If P_2 is $4P_1$ and l_2 is $\frac{l_1}{4}$, then the radius r_2 will be equal to:

1) r_1 2) $2r_1$ 3) $4r_1$ 4) $\frac{r_1}{2}$

9. Statement I : When a Si sample is doped with Boron, it becomes P type and when doped by Arsenic it becomes N-type semi conductor such that $P-$ type has excess holes and N-type has excess electrons.

Statement II : When such P-type and N-type semiconductors are fused to make a junction, a current will automatically flow which can be detected with an externally connected ammeter. In the light of above statements, choose the most appropriate answer from the options given below.

1)Statement I is true and Statement II is false

2)Statement I is false and Statement II is true

3)Statement I and statement II are true

4)Statement I and statement II are false

10. A simple telescope, consisting of an objective of focal length 60 cm and a single eye lens of focal length 5 cm is focused on a distant object in such a way that parallel rays emerge from the eye lens. If the object subtends an angle of 2° at the objective, the angular width of the image is

1) 10^{0} 2) 24^{0} 3) 50° 4) $(1/6)^{0}$

11. Pick out the correct statements

i) Susceptibility of a diamagnetic substance is high and negative

ii) In paramagnetic substance, the intrinsic magnetic moment is not zero

iii) When a paramagnetic substance is heated, it becomes ferromagnetic

iv) Ferro magnetic material becomes paramagnetic above a certain temperature

1) i and iii   2)iii and iv   3)ii and iii   4)ii and iv

12. Given below are two statements:

Statement I : For diamagnetic substance $-1 \le \chi < 0$, where χ is the magnetic susceptibility.

Statement II : Diamagnetic substances when placed in an external magnetic field, tend to move from stronger to weaker part of the field.

In the light of the above statements, choose the correct answer from the options given below.

1)Both statements I and II are true

2)Statement I is true but statement II is false

3)Statement I and statement II are false

4)Statement I is false but statement II is true

13 A cube of each side r has edges parallel to x, y and z-axis of rectangular coordinate system. During the propagation of electromagnetic wave the electric field E is parallel to y-axis and the magnetic field is parallel to x-axis. The rate at which energy flows through each face of the cube is (E and B are instantaneous values)

1) $\frac{r^2EB}{2\mu_0}$ in parallel to $x-y$ plane face and 0 in others

2) $\frac{r^2EB}{\mu_0}$ in parallel to $x-y$ plane face and 0 in others

3) $\frac{r^2EB}{\mu_0}$ in all faces

4) $\frac{r^2EB}{\mu_0}$ in parallel to $y-z$ plane face and 0 in others

14. This question has Statement - I and Statement - II. Of the four choices given after the Statements, choose the one that best describes the two Statements.

Statement I: Very large size telescopes are reflecting telescopes instead of refracting telescopes.

Statement II: It is easier to provide mechanical support to large size mirrors than large size lenses.

1)Statement I is true and Statement II is false

2)Statement I is false and Statement II is true

3)Statement I and statement II are true and statement II is correct explanation for statement I

4)Statement I and statement II are true and statement II is not the correct explanation for statement I

15. Match List I with List II:

List - I		List - II	
A	Intrinsic semiconductor	I	Fermi-level near the valence band
B	n-type semiconductor	II	Fermi-level in the middle of valence and conduction band
C	p-type semiconductor	III	Fermi-level near the conduction band
D	Metals	IV	Fermi-level inside the conduction band

Choose the correct answer from the options given below:

1) $A-III, B-I, C-IV, D-II$ 2) $A-I, B-II, C-III, D-IV$

3) $A-II, B-III, C-I, D-IV$ 4) $A-II, B-I, C-III, D-IV$

16. In Fraunhoffer diffraction at single slit of width d with incident light of wavelength $5500A^0$, the first minimum is observed at angle of 30°. The first secondary maximum is observed at an angle θ, equal to

1) $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$ 2) $\sin^{-1}\left(\frac{1}{4}\right)$ 3) $\sin^{-1}\left(\frac{3}{4}\right)$ 4) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$

17. Which of the following is incorrect about a simple microscope?

1)It forms erect and real image

2)It forms erect and virtual image

3)The object is kept at a distance of less than one focal length from the lens

4)It brings the object closer to the eye than near point

18. An iron rod is placed coaxially inside a solenoid on which the number of turns per unit length is 600 . The relative permeability of the rod is 1000 . If a current of $0.5\ A$ is passed through the solenoid, then the magnetization of the rod will be

1) $2.997\times10^{2}\ A/m$ 2) $2.997\times10^{3}\ A/m$

3) $2.997\times10^{4}\ A/m$ 4) $2.997\times10^{5}\ A/m$

19 A single slit of width b is illuminated by a coherent monochromatic light of wavelength λ . If the second and fourth minima in the diffraction pattern at a distance $1\ m$ from the slit are at $3\ cm$ and $6\ cm$ respectively from the central maximum, what is the width of the central maximum? (i.e. distance between first minimum on either side of the central maximum)

1) 1.5 cm 2) 3.0 cm 3) 4.5 cm 4) 6.0 cm

20 The length, breadth and mass of two bar magnets are same but their magnetic moments are $3\ M$ and $2M$ respectively. These are joined in parallel with similar poles on same side and are suspended by a string. When oscillated in a magnetic field of strength B, the time period obtained is $5\ s$. If the poles of either of the magnets are reversed then the time period of the combination in the same magnetic field will be

1) $2\sqrt{2}s$ 2) $5\sqrt{5}s$ 3) $1\ s$ 4) $3\sqrt{3}s$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:
Marking scheme: +4 for correct answer, -1 in all other cases.

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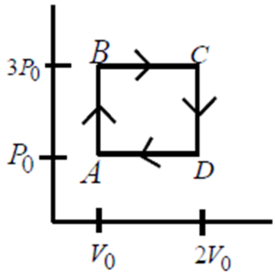
21. A point source of electromagnetic radiation has an average power output of $800\ W$. The maximum value of electric field at a distance $4.0\ m$ from the source in $\frac{V}{m}$ is $\sqrt{1000K}$. Find the value of K approximated to nearest integer.

22. Unpolarized light of intensity $64\ W\ m^{-2}$ passes through three polarizers such that the transmission axis of the last polarizer is crossed with that of the first. The intensity of final emerging light is $3\ W\ m^{-2}$. The intensity of light transmitted by first polarizer will be

23. An engine operates by taking a monatomic ideal gas through the cycle shown in the figure. The percentage efficiency of the engine is $\frac{400}{K}$. Find the value of K.

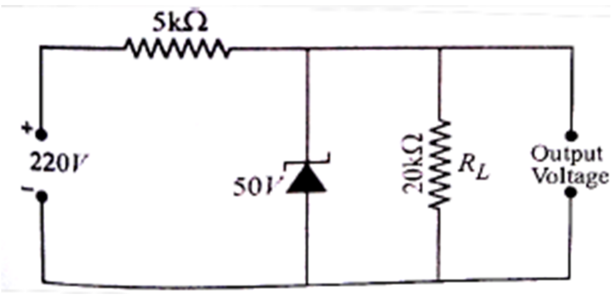


24. A microscope has an objective of focal length 1.5 cm and an eye piece of 2.5 cm. If the distance between objective and eye piece is 25 cm, is the value of magnification produced for relaxed eye is $50n$. Find the value of n approximated to nearest integer.

25. Two magnets are suspended by a given wire one by one in a horizontal uniform magnetic field. In order to deflect the first magnet through 45°, the wire has to be twisted through 540° whereas with the second magnet, the wire requires a twist of 360° for the same deflection. Then the magnetic moments of the first and second magnets are in the ration:14. Find the value of n

26. A single slit of width a is illuminated by violet light of wavelength 400 nm and the width of the primary diffraction maximum is measured as y. When half of the slit width is covered and illuminated by red light of wavelength 700 nm, the width of the primary diffraction maximum $\frac{ny}{4}$. Find the value of n

27 For the zener diode circuit shown, the current through the zener in mA is I. Find $2I$



28. A parallel plate capacitor made of circular plates each of radius $R = 6$ cm has capacitance $C = 100 pF$. The capacitor is connected to a 230 V a.c. supply with an angular frequency of $300\ s^{-1}$. The rms value of displacement current in μA is I. Find the value of $10I$.

29. An N - type silicon sample of length 6×10^{-2} m has width 4×10^{-3} m and thickness 25×10^{-5} m. When a voltage is applied across the length of the sample, the current in sample is 4.8 mA. If the free electron density in sample is 10^{22} m^{-3}, then the time taken for the electrons to travel the full length of the sample in seconds is t. Find $400t$.

30. A person cannot see the objects beyond 100 cm. The focal length of the lens in cm required to correct this vision is x. Find the value of $\frac{x^2}{25}$.

CHEMISTR **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31 Metallic bond is a bond between

1)Valence electron and positively charged metal ions

2)The ions of two different metals

3)a metal and non metal

4)Two metal atoms

32. Match the weight of reactants given in Column I with weight of products marked (?) given Column II

	List - I		List - II
A	$2H_2+O_2\rightarrow 2H_2O$ 1.0gm1.0gm?	p	0.56gm
B	$N_2+3H_2\rightarrow 2NH_3$ 1.0gm1.0gm ?	q	1.333gm
C	$CaCO_3\rightarrow CaO+CO_2$ 1.0gm	r	1.125gm
D	$C+2H_2\rightarrow CH_4$ 1 g1 g ?	s	1.214gm

1) A – q, B – p, C – r, D – s 2) A – s, B – p, C – q, D – r

3) A – r, B – s, C – p, D – q 4) A – r, B – q, C – s, D – p

33 If λ_0 and λ be the threshold wavelength and the wavelength of incident light, the speed of photo-electrons ejected from the metal surface is:

1) $\sqrt{\frac{2h}{m}(\lambda_0-\lambda)}$ 2) $\sqrt{\frac{2hc}{m}(\lambda_0-\lambda)}$

3) $\sqrt{\frac{2hc}{m}\left(\frac{\lambda_0-\lambda}{\lambda\lambda_0}\right)}$ 4) $\sqrt{\frac{2h}{m}\left(\frac{1}{\lambda_0}-\frac{1}{\lambda}\right)}$

34. Aqueous solution of a salt (Y) is alkaline to lithmus. On strong heating, it swells-up to give a glassy material. When conc. H_2SO_4 is added to a hot concentrated solution of (Y), while crystals of a weak acid separate out. Hence, the compound (Y) is

1) $Na_2SO_4 \cdot 10H_2O$ 2) $Ca_2P_6O_{11} \cdot 10H_2O$

3) $Na_2\ B_6O_{11}$ 4) $Na_2\ B_4O_7 \cdot 10H_2O$

35. Statement I : For the physical equilibrium $H_2O(s) \rightleftharpoons H_2O(l)$ on increasing temperature and increasing pressure more water will form.

Statement II: Since forward reaction is endothermic in nature and volume of water is greater than that of the volume of ice.

1)If both the statements are TRUE and statement II is the correct explanation of statement I

2)If both the statements are TRUE but statement II is not the correct explanation of statement I

3)If statement I is TRUE and statement II is FALSE

4)If statement I is FALSE and statement II is TRUE

36. Statement **–I** : D.D.T is a non-biodegradable pollutant

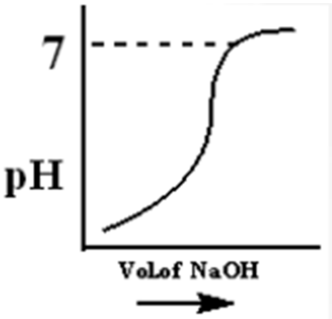
Statement - II : D.D.T insecticide used in agriculture

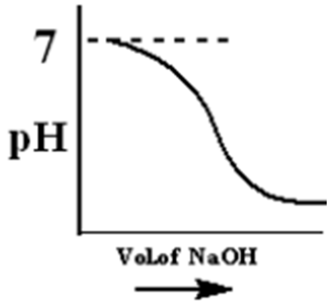
1)Both statements are correct 2)Both statements are wrong

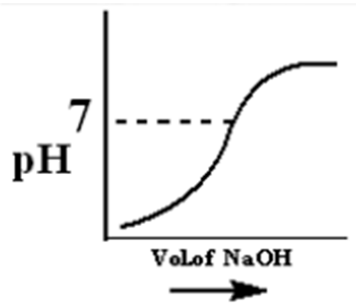
3)Statement I is false but statement II is true

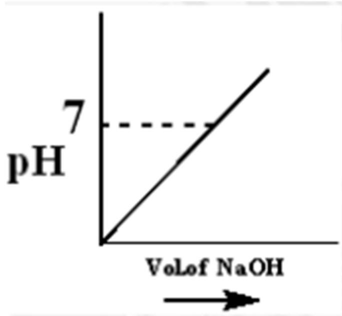
4)Statement I is true statement II is false

37. 100 mL of $0.1\text{M}\,HCl$ is taken in a beaker and to it 100 mL of $0.1\text{M}\,NaOH$ is added in steps of 2 mL and the pH is continuously measured. Which of the following graphs correctly depicts in change in pH ?

1) 

2) 

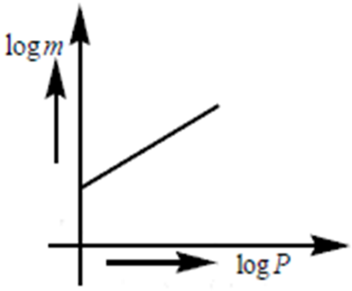
3) 

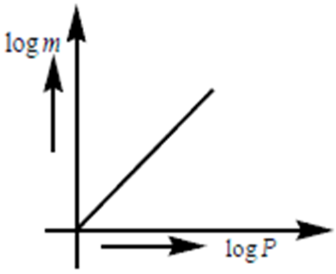
4) 

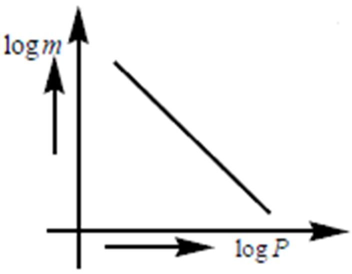
38. Rutherford's experiment on scattering of alpha particles showed for the first time that atom has:

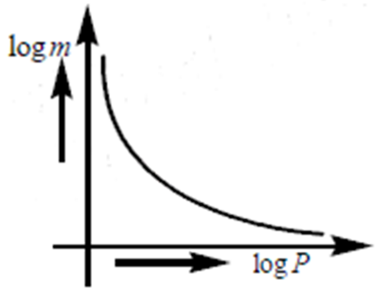
1)electrons 2)protons 3)nucleus 4)neutrons

39. Which of the following curves represents the Henry's law?

1) 

2) 

3) 

4) 

40. The Lassaigne's extract is boiled with dil HNO_3 before testing for halogens because

1) $AgCN$ is soluble in HNO_3

2)Silver halide are soluble in HNO_3

3) $NaCN$ and Na_2S are decomposed by HNO_3

4) Ag_2S is soluble in HNO_3

41. An octahedral complex with molecular composition $M.5NH_3.Cl.SO_4$ has two isomers, A and B. The solution of A gives a white precipitate with $AgNO_3$ solution and the solution of B gives white precipitate with $BaCl_2$ solution. The type of isomerism exhibited by the complex is:

1)Linkage isomerism 2)Ionisation isomerism

3)Coordinate isomerism 4)Geometrical isomerism

42.

$$\underset{\text{Sucrose}}{C_{12}H_{22}O_{11}} + H_2O \xrightarrow{EnzymeA} \underset{Glucose}{C_6H_{12}O_6} + \underset{Fructose}{C_6H_{12}O_6}$$

$$\underset{Glucose}{C_6H_{12}O_6} \xrightarrow{EnzymeB} 2C_2H_5OH + 2CO_2$$

In the above reactions, the enzyme A and enzyme B is respectively are

1)Amylase and Invertase 2)Invertase and Zymase

3)Zymase and Invertase 4)Invertase and Amylase

43 For lead storage battery pick the correct statements

A) During charging of battery, $PbSO_4$ on anode is converted into PbO_2

B) During charging of battery, $PbSO_4$ on cathode is converted into PbO_2

C) Lead storage battery, consists of grid of lead packed with PbO_2 as anode

D) Lead storage battery has $\sim 38\%$ solution of sulphuric acid as an electrolyte Choose the correct answer from the options given below:

1)B, D only 2)B, C, D only 3)A, B, D only 4)B, C only

44. Two oxides of metal contains 50% and 40% of metal M respectively. If the formula of the first oxide is MO_2. Then formula of the second oxide will be:

1) M_2O_5 2) MO_3 3) M_2O 4) MO_2

45 ***Assertion:*** Thin layer chromatography is an adsorption chromatography.

Reason: A thin layer of silica gel is spread over a glass plate of suitable size in thin layer chromatography which acts as an adsorbent.

In the light of the above statements, choose the correct answer from the options given below:

1)Both A and R are true and R is the correct explanation of A

2)Both A and R are true but R is NOT the correct explanation of A

3) A is true but R is false

4) A is false but R is true

46. Which of the following used to reagent is used to test for phosphate ion is?

1)Nessler's reagent   2)Ammonium molybdate

3)Baeyer's reagent   4)Fentos reagent

47. Match List - I with List - II

List - I (Mixture)		List - II (Purification Process)	
A	Chloroform and Aniline	I	Steam distillation
B	Benzoic acid and Naphthalene	II	Sublimation
C	Water and Aniline	III	Distillation
D	Naphthalene and Sodium chloride	IV	Crystallization

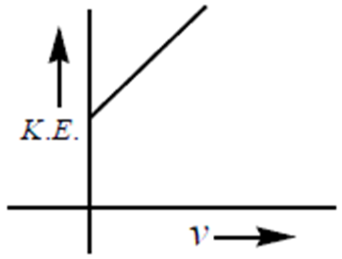
1) $A-IV, B-III, C-I, D-II$ 2) $A-III, B-I, C-IV, D-II$

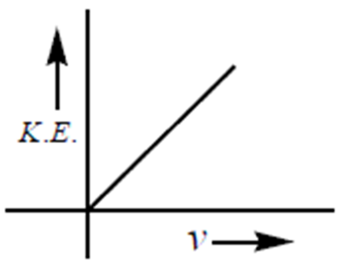
3) $A-III, B-IV, C-II, D-I$ 4) $A-III, B-IV, C-I, D-II$

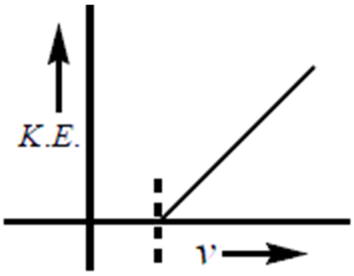
48. Which of the vitamins the one whose deficiency causes rickets (bone deficiency) is

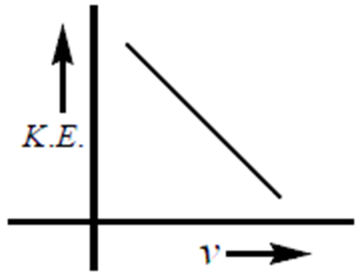
1)Vitamin E 2)Vitamin B 3)Vitamin D 4)Vitamin C

49. According to Einstein's photoelectric equation, the graph between kinetic energy of photoelectrons ejected and the frequency of the incident radiation is:

1) 

2) 

3) 

4) 

50. The compound formed in the borax bead test of Cu^{2+} ion is oidising flame is

1) Cu 2) $CuBO$ 3) $Cu(BO_2)_2$ 4)None of these

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

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51. The work function of sodium metal is 4.41×10^{-19} J. If photons of wavelength 300 nm are incident on the metal, the kinetic energy of the ejected electrons will be

$(h=6.63\times10^{-34} J_s; c=3\times10^{8}\ m/s)$ _______ $\times10^{-21}$ J.

52. During estimation of Nitrogen present in an organic compound by KJeldahl's method, the ammonia evolved from 0.5gm of the compound in KJeldahl's estimation of nitrogen, neutralized 10ml of $1MH_2SO_4$. The percentage of nitrogen in the compound is x. What is x?

53. Two elements A and B which form 0.15 moles of A_2B and AB_3 type compounds. if both A_2B and AB_3 weigh equally, then the atomic weight of A is times of atomic weight of B.

54. Henry's law constant for CO_2 is water is 1.6×10^{8} Pa at 298 K. The quantity of CO_2 is 500 ml of soda water when packed under 2.5 atmCO_2 pressure at 298 K is xgm. Then x is (Round off the nearest Integer).

55. $CH_3-C\equiv CH-\frac{\text{Red hot}}{2000^{\circ}Cu}Cu$, the no of π bonds in X is

56. The number of $4f$ electrons in the ground state electronic configuration of Gd^{2+} is [Atomic number of $Gd = 64$]

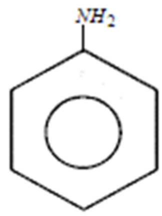
57. A $5.0ml$ solution of H_2O_2 liberated $1.27gm$ of iodine from an acidified KI solution. The percentage strength of H_2O_2 is x. The value of x is (Round of the nearest integer)

58. H_3A is a weak tri basic acid with $Ka_1 = 10^{-5}, Ka_2 = 10^{-9}$ and $Ka_3 = 10^{-13}$. The value of P^X of $0.1MH_3\ A$ solution.

Where $P^X = -\log_{10} X$ and

$X = \frac{[A^{-3}]}{[HA^{-2}]}$ is

59.. Amongest the following the total number of compounds which does not give Lassaigne's test for nitrogen

i) 

ii) $H_2\ N-CH_2-COOH$

iii) $H_2\ N-NH_2$

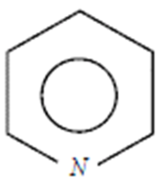
iv) $H_2\ N-OH$

v) Urea

vi) $PhNO_2$

vii) CH_3CHO

viii) CH_3CN

ix) 

60. In carius method of estimation of halogen's 250mg of an organic compound gave 141mg of $AgBr$. The percentage is x. Then x is

MATHS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. A random variable X has the following probability distribution:

$X:$	1	2	3	4	5	6	7	8
$p(X):$	0.15	0.23	0.12	0.10	0.20	0.08	0.07	0.05

For the events $E=\{X \text{ is a prime number }\}$ and $F=\{X<4\}$, then the value of $100P(E\cup F)$ is

1)87 2)77 3)35 4)50

62. Let a set $A=A_1\cup A_2\cup\ldots\ldots\cup A_k$, where $A_i\cap A_j=\phi$ for $i\neq j, 1\leq i, j\leq k$. Define the relation R from A to A by $R=\{(x,y): y\in A_i \text{ if and only if } x\in A_i, 1\leq i\leq k\}$. Then, R is:

1)reflexive, symmetric but not transitive

2)reflexive, transitive but not symmetric

3)reflexive but not symmetric and transitive

4)reflexive, symmetric and transitive

63. Consider the following frequency distribution:

Class:	$0-6$	$6-12$	$12-18$	$18-24$	$24-30$
Frequency :	a	b	12	9	5

If mean $\frac{309}{22}$ and median $=14$, then the value $2a+3b$ is equal to

1) 8 2)36 3)46 4)10

64. In a class, 18 students offered physics, 23 offered chemistry and 24 offered Mathematics. Of these, 13 are in both chemistry and Mathematics, 12 in Physics and Chemistry, 11 in Physics and Mathematics, and 6 in all the three subjects.

Statement I : The number of students offered Mathematics but not Chemistry $=11$

Statement II : The number of students who offered exactly one of the three subjects $=11$

1)Statement I is true and Statement II is false

2)Statement I is false and Statement II is true

3)Statement I and statement II both are true

4)Statement I and statement II both are false

65. An organization awarded 48 medals in event ' A ', 25 in event ' B ' and 18 in event ' C '. If these medals went to total 60 men and only five men got medals in all the three events, then, the number of men received medals in exactly two of three events is k then $10k$ is

1)100 2)90 3)210 4)150

66. Statement I : $R_1 = \{(a,b) \in N \times N : |a-b| \le 13\}$ is an equivalence relation

Statement II : $R_2 = \{(a,b) \in N \times N : |a-b| \ne 13\}$. is an equivalence relation

1)Statement I is true and Statement II is false

2)Statement I is false and Statement II is true

3)Statement I and statement II both are true

4)Statement I and statement II both are false

67. Two finite sets have m and n elements. The total number of subsets of the first set is 56 more than the total number of subsets of the second set. The value of $m+n$ is

1) 9 2)6 3)3 4)18

68 If R be a relation ' $<$ ' from $A = \{1,2,3,4\}$ to $B = \{1,3,5\}$, i.e., $(a,b) \in R \Leftrightarrow a < b$, then RoR^{-1} is

1) $\{(1,3),(1,5),(2,3),(2,5),(3,5),(4,5)\}$

2) $\{(3,1),(5,1),(3,2),(5,2),(5,3),(5,4)\}$

3) $\{(3,3),(3,5),(5,3),(5,5)\}$

4) $\{(3,3),(3,4),(4,5)\}$

69. If for some $x \in R$, the frequency distribution of the marks obtained by 20 students in a test is

Marks	2	3	5	7
Frequency	$(x+1)^2$	$2x-5$	x^2-3x	x

Then 20 times of the mean of the marks is:

1)56 2)30 3)25 4)32

70. A random variable X has the following probability distribution:

$X=x$	0	1	2	3	4	5	6	7	8
$P(X=x)$	a	3a	5a	7a	9a	11a	13a	15a	17a

Statement I: The value of ' a ' is $\frac{1}{81}$

Statement II: Mean of random variable X is 6

1)Statement I is true and Statement II is false

2)Statement I is false and Statement II is true

3)Statement I and statement II both are true

4)Statement I and statement II both are false

71. A random variable x takes the values $0,1,2,3$ and its mean is 1.3. If $P(x=3)=2P(x=1)$ and $P(x=2)=0.3$, then $P(x=0)+P(x=1)$ is

1)0.4 2)0.1 3)0.5 4)0.2

72. Let R and S be two relations on a set A. Consider the following statements: $S_1:R$ and S are transitive, then $R\cup S$ is also transitive $S_2:R$ and S are transitive, then $R\cap S$ is also transitive $S_3:R$ and S are reflexive, then $R\cap S$ is also reflexive $S_4:R$ and S are symmetric then $R\cup S$ is also symmetric Then correct statements are:

1) S_1,S_2,S_4 2) S_1,S_2,S_3,S_4 3) S_1,S_2,S_3 4) S_2,S_3,S_4

73. Let the mean of 6 observation $1,2,4,5,x$ and y be 5 and their variance be 10. Then three times of the mean deviation about the mean of the given data is equal to

1) $\frac{8}{3}$ 2) $\frac{7}{3}$ 3)3 4)8

74. The group of beautiful girls is

1)A null set 2)A finite set

3)An infinite set 4)Not a set

75. Let μ be the mean and σ be the standard deviation of the distribution.

x_i	0	1	2	3	4	5
f_i	$k+2$	$2k$	k^2-1	k^2-1	k^2+1	$k-3$

Where $\Sigma f_i = 62$.

Statement I : Value of $k = 3$

Statement II : $5\left[\mu^2+\sigma^2\right]=40$ where $[x]$ denote the greatest integer $\le x$,

1)Statement I is true and Statement II is false

2)Statement I is false and Statement II is true

3)Statement I and statement II both are true

4)Statement I and statement II both are false

76. For a biased dice, the probabilities for the different faces to turn up are given below:

X	1	2	3	4	5	6
$P(X)$	0.10	0.32	0.21	0.15	0.05	0.17

Then the mean of the above distribution is:

1)3.24 2)3.34 3)4.24 4)4.34

77. Let R be an equivalence relation on a finite set A having n elements. Then the number of ordered pairs in R is:

1)less than n 2)greater than or equal to n

3)less than or equal to n 4)only one

78. The range of a random variable $x = \{1, 2, 3 \ldots\ldots\}$ and the probability are given by

$P(x=k) = \frac{{}_3CK}{K!}(k=1,2,3\ldots.)$ and C is a constant. Then $2C =$

1) $\log_2(\log 3)$

2) $\frac{1}{2}\log(\log 2)$

3) $\frac{\log_e(\log 2)}{\log_3^e}$

4) $2\log_3(\log 2)$

79. The mean and variance of a set of 15 numbers are 12 and 14 respectively. The mean and variance of another set of 15 numbers are 14 and σ^2 respectively. If the variance of all the 30 numbers in the two sets is 13 .

Statement I : Value of $\sigma^2 = 10$

Statement II : Combined variance $= \sigma^2 = \dfrac{n_1\sigma_1^2 + n_2\sigma_2^2}{n_1 + n_2} + \dfrac{1}{(n_1+n_2)^2}\left(\bar{x}_1 - \bar{x}_2\right)^2$

1) Statement I is true and Statement II is false

2) Statement I is false and Statement II is true

3) Statement I and statement II both are true

4) Statement I and statement II both are false

80. Let $A = \{x \in R : |x+1| < 2\}$ and $B = \{x \in R : |x-1| \geq 2\}$. Then which one of the following statements is true?

1) $A - B = (-3, 1)$

2) $B - A = R - (-3, 3]$

3) $A \cap B = (-3, -1]$

4) $A \cup B = R - [1, 3]$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. In a moderately skewed distribution, the values of mean and median are 5 and 6 respectively. The value of mode for such distribution is k then k^2 is

82. Let $A = \{n \in N : n$ is a 3-digit number]

$B = \{9k + 2 : k \in N\}$ And $C = \{9k + \ell : k \in N\}$ for some $\ell (0 < \ell < 9)$

If the sum of all the elements of the set $A \cap (B \cup C)$ is 274×400, then 3ℓ is equal t

83. Let A be set of first ten natural numbers and R be a relation on A, defined by $(x, y) \in R \Rightarrow x + 2y = 10$. Then number of elements in range of R is

84. The probability distribution of the random variable X is as follows:

X	-3	-2	-1	0	1	2	3
$P(X)$	5k	3k	k	k	k	2k	k

If 2^{nd} moment about origin of the random variable X is P, then $14P$ is

85. The marks of some students were listed out of 75. The SD of marks was found to be 9. Subsequently the marks were raised to a maximum of 100 and variance of new marks was calculate. Then twice the new variance is:

86. Let $S=\{4,6,9\}$ and $T=\{9,10,11,\ldots\ldots,1000\}$. If $A=\{a_1+a_2+\ldots\ldots+a_k : k\in N, a_1, a_2, a_3, \ldots\ldots, a_k \in S\}$, then twice the sum of all the elements in the set $T-A$ is equal to

87. The probability distribution of X is:

X	0	1	2	3
$P(X)$	$\frac{1-d}{4}$	$\frac{1+2d}{4}$	$\frac{1-4d}{4}$	$\frac{1+3d}{4}$

For the minimum possible value of $d, 120$ times the mean of X is equal to

88. Let R_1 and R_2 be relations on the set $\{1,2,\ldots\ldots\ldots,50\}$ such that $R_1=\{(p,p^n) : p$ is a prime and $n\geq 0$ is an integer $\}$ and $R_2=\{(p,p^n) : p$ is a prime and $n=0$ or $1\}$. Then 3 times the number of elements in R_1-R_2 is

89.. If $A=\{1,2,3\}$, the number of reflexive relation in A is k then $\frac{k}{4}$ is

90. In a class of 100 students there are 70 boys whose average marks in a subject is 75 . If the average marks of the complete class is 72 , and the average marks of the girls is x then $\frac{x}{5}$ is